

HKBK College of Engineering

Nagavara, Bengaluru – 560045

Major Project phase -1 (BEC786)

Department : Electronics and Communication Engineering

Project Title: “IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation”



Batch No: 28

Students name:

Kirankumar Goudar - 1HK22EC072

Melwin Jain - 1HK22EC088

Mutturaj - 1HK22EC104

Nithin Sadashiv Shipure - 1HK22EC117

Guide Name

Prof. Shruthi B M

Assistant Professor

Dept. of ECE

Introduction:

- The Smart Bat System is an IoT-based training bat that gives real-time feedback on shot timing, impact location, and stroke direction using embedded sensors and wireless connectivity.
- It is designed to overcome the limitations of traditional coaching, which relies on subjective observation and cannot capture micro-errors in bat angle, timing, or body movement.
- By combining low-cost IoT hardware with sensor fusion, the project brings advanced analytics that are usually limited to professional setups to grassroots and academy-level players.

Objectives:

- Provide real-time, objective feedback on batting technique by detecting impact location (top edge, middle, bottom) and classifying strokes (forward, right, left).
- Use motion sensing (MPU6050) to analyze bat orientation and swing path for different shots and guide players through focused training modes via Blynk virtual buttons.
- Offer an affordable, portable, and consistent training tool that democratizes access to data-driven cricket coaching for beginners, amateurs, and academies.
- To minimize learning time for beginners by converting subjective observations into clear messages such as “Perfect Shot,” “Right Cover Drive,” “Left Drive,” and “Bad Shot.”

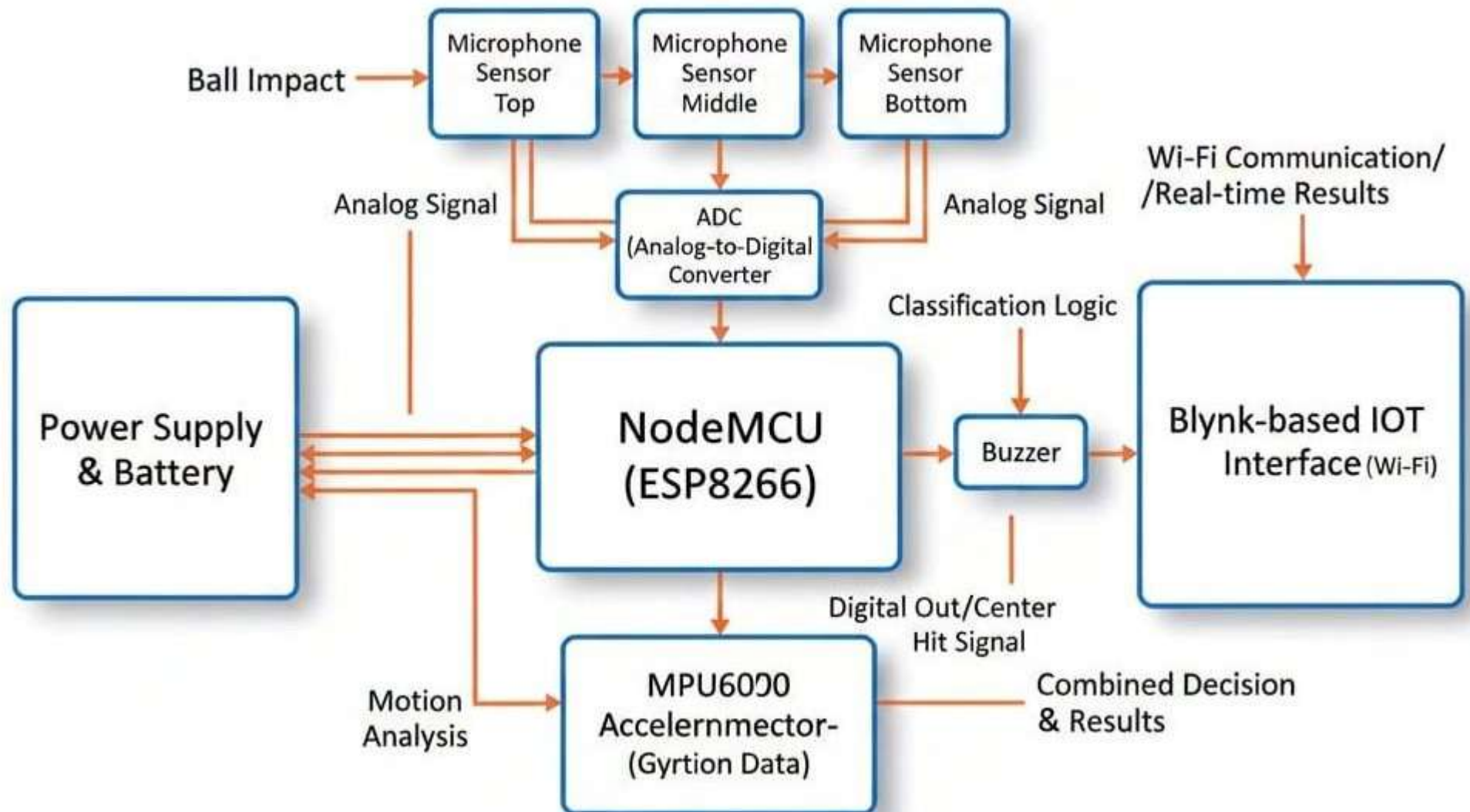
Literature Survey:

Authors	Year	Title / Project Description	Methods Used	Drawbacks / Limitations
Kumar et al.	2020	Designed a Wearable Cricket Stroke Analyzer using *MPU6050* to detect bat swing, orientation and stroke types. Data transmitted to mobile via Bluetooth.	IMU-based angle measurement, accelerometer & gyroscope features, threshold-based classification.	Limited accuracy due to simple thresholds, no real-time web dashboard, Bluetooth range very short, no impact detection.
Singh & Reddy	2021	Developed a Smart Cricket Bat with *vibration sensors* to detect sweet spot vs edge hits. Provided stroke quality feedback via mobile app.	Piezo vibration sensor, FFT for vibration classification, Android app.	Only detects sweet spot/edge, cannot classify shot type, no cloud/web connectivity, poor performance outdoors.
Patel et al	2022	Built a cricket training system using *ESP8266 + MPU6050* to detect	IMU + Wi-Fi + Blynk IoT dashboard,	Blynk has limited customisation, no proper data

		swing speed, bat path, and ball impact. Used Blynk app for displaying data.	moving average filter.	logging, misclassification due to noisy IMU, cannot track multiple sessions.
Rao & Deshmukh	2023	Proposed a Real-Time Cricket Shot Classification System using IMU + ML (SVM classifier). Classified straight, cover, pull, and sweep.	ML on IMU features (angular velocity, RMS acceleration).	Requires large training dataset, heavy computation → not suitable for ESP8266, latency issues, offline model only.
Sharma & Gupta	2024	Built a Web Dashboard for Sports Motion Analysis using WebSockets to stream sensor data to browser. Demonstrated with bat swing prototype.	Node.js WebSocket server, real-time graphs on browser, MPU-based hardware.	No shot detection logic, only raw data streaming; required strong Wi-Fi; not usable outdoors without server.

System Design / Methodology:

Block Diagram:



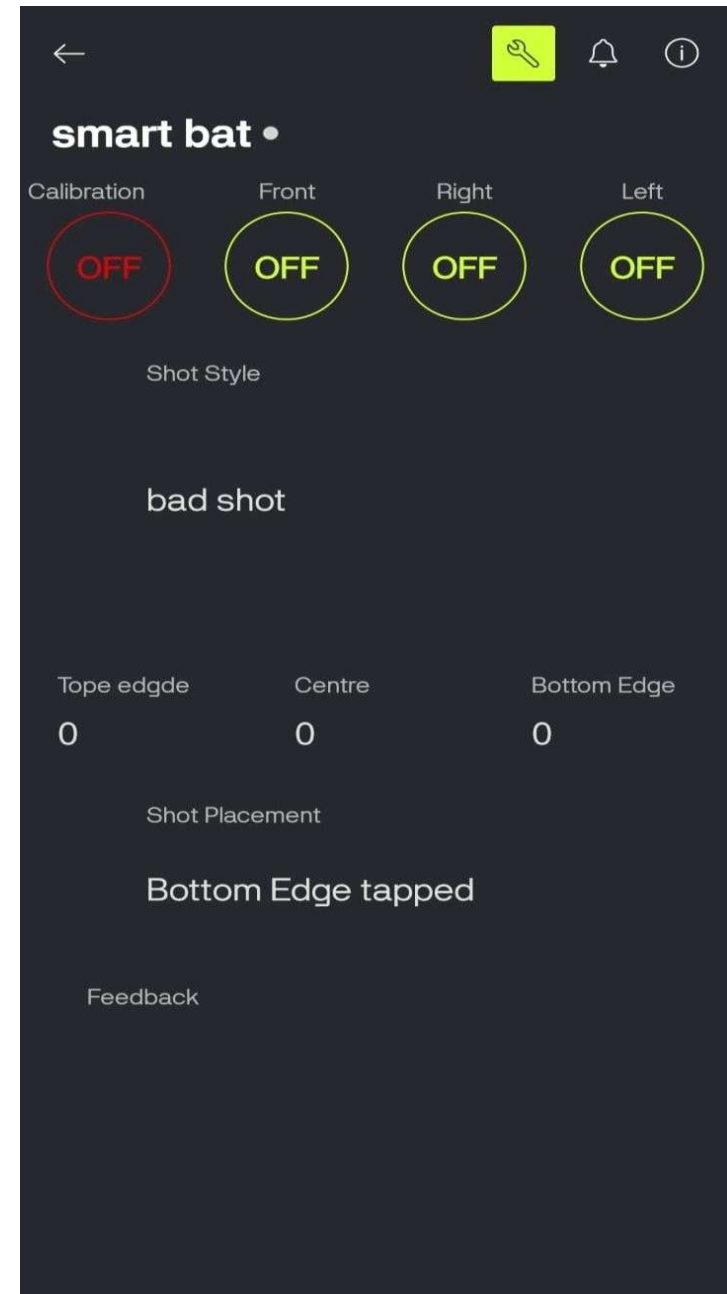
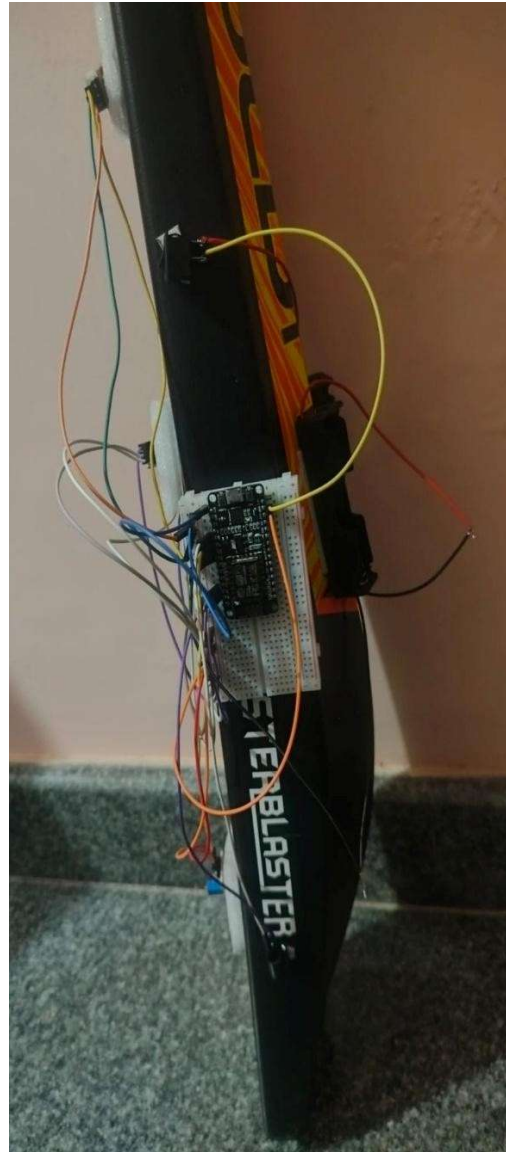
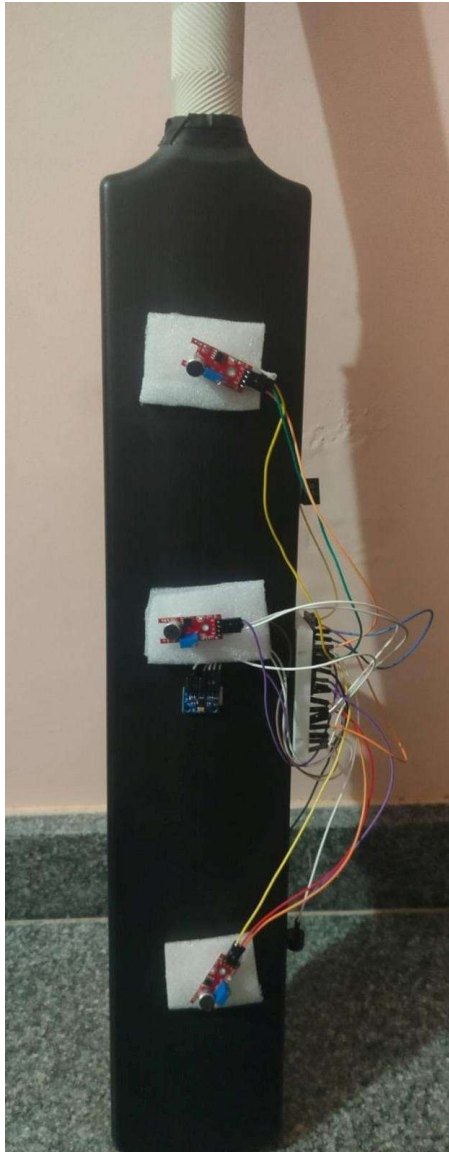
Methodology:

1. Initialize Node MCU, mic sensors, MPU6050 and buzzer, and connect the bat to the Blynk mobile app over Wi-Fi.
2. Perform calibration once by holding the bat in rest position so baseline MPU6050 readings are stored as reference.
3. Continuously read filtered mic and motion data, detect ball impact, and classify it as top edge, center or bottom edge, triggering the buzzer for center hits.
4. Let the player select a single mode (forward, right or left) and compare live MPU6050 readings with predefined motion patterns to label each stroke as correct or bad.
5. Send final feedback (impact type, mode and correctness) instantly to the Blynk dashboard for the player and coach.

Implementation:

- Firmware on Node MCU initializes sensors (I²C for MPU6050, ADC for mics), implements threshold-based impact detection, motion pattern matching, mode-conflict handling, and Blynk.virtualWrite communication.
- Hardware is housed in a shock-resistant enclosure mounted on the bat, with embedded mic strips along the blade and a rechargeable Li-ion/Li-Po supply regulated to 3.3 V for long practice sessions.

Results / Sample Output:



Applications:

- Usable in cricket academies, schools, and clubs for objective player evaluation, structured drills, and efficient monitoring of multiple students simultaneously.
- Supports self-training, remote coaching, sports-science research, and IoT/embedded-systems education by providing live motion and impact data over the cloud.
- Can be used by individual players practicing in small grounds or indoor spaces as a personal digital coach that gives real-time feedback without needing a coach always present.
- Helpful for schools, clubs and academies that want consistent, data-driven evaluation of many students, reducing dependence on the subjective judgment of different coaches.

Challenges Faced:

- Ensuring reliable vibration detection without false triggers required careful threshold selection, sensor placement, and mechanical isolation against bat shocks and ambient noise.
- Maintaining stable Wi-Fi connectivity, low latency, compact weight distribution, and adequate battery life on a vibrating, moving bat demanded iterative hardware integration and power-management tuning.

Future works:

- Add advanced analytics such as swing speed, impact force, long-term performance tracking, and AI-based scoring or stroke-fault detection using stored sensor data.
- Integrate with additional smart equipment (gloves, shoes, stumps, ball) and AR/VR or video-synchronization features to build a complete smart-cricket training ecosystem.

Conclusion:

- The Smart Bat System proves that low-cost sensors and IoT can convert a normal cricket bat into a smart, real-time coaching device that improves timing, technique, and training efficiency.
- By delivering instant, unbiased feedback and supporting scalable enhancements, the project offers a practical step toward technology-driven, data-centric cricket coaching at all levels.

References:

[1] Deshmukh, P. & Kumar, S.

Design and Implementation of a Smart Cricket Bat with Integrated Sensors and Wireless Data Transmission, IJSRCSEIT, Vol. 8, Issue 5, pp. 133–145.

[2] Agarwal, P. & D'Souza, S. N.

Cricket Bat Swing Detection Based on 6-Axis IMU, BMS Institute of Technology, Bengaluru, 2025 (Preprint / Report).

Author: *Piyush Agarwal*.

[3] Morishita, Y. & Jinji, T.

Accuracy and Error Trends of Commercially Available Bat Swing Sensors in Baseball, Sports, 2022.

Source: *PMC*.

[4] Sharma, M., Anand, A., Srivastava, R., & Kaligounder, L.

Wearable Audio and IMU Based Shot Detection in Racquet Sports, arXiv, 2018.

[5] Fuss, F. K., Doljin, B., & Ferdinands, R. E. D.

Mobile Computing with a Smart Cricket Ball: Discovery of Novel Performance Parameters and Their Practical Application, Sensors, 2021.

